Roll No.: | | | | | | | | | | | |

Test Date: 20-07-2025

Aakash
Medical | IIT-JEE | Foundations

A
Booklet Code



A I A T S
ALL INDIA AAKASH TEST SERIES

Aakash Rajasthan
Goswami Ji

for

Medical Entrance Exam - 2026

National Eligibility-cum-Entrance Test (NEET)

TEST No. 1

(XII Studying Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 1

TOPICS OF THE TEST

Physics	Electric Charges and Fields, Electrostatic Potential and Capacitance
Chemistry	Solutions, Electrochemistry
Botany	Sexual Reproduction in Flowering Plants
Zoology	Human Reproduction

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Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

All India Aakash Test Series for NEET-2026

Test - I

(XII Studying Students)

[PHYSICS]

Choose the correct answer:

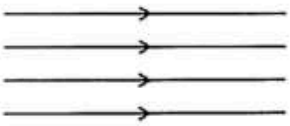
1. A 16 μC point charge is divided into two charges such that when they are placed 10 cm in vacuum experiences maximum coulombic force. That maximum force will be

   (1) 43.2 N (2) 57.6 N

   (3) 54 N (4) 50 N

2. Electric field lines in a certain region of space are shown in figure.

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   Then, choose the correct option.

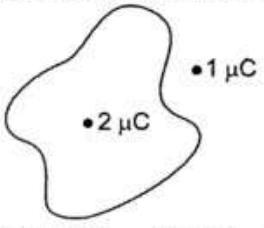
   (1) Both magnitude and direction of electric field are uniform

   (2) Both magnitude and direction of electric field are non-uniform

   (3) Only magnitude of electric field is uniform

   (4) Only direction of electric field is uniform

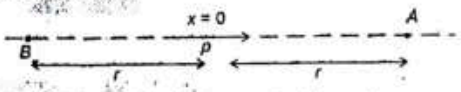
3. Total flux through Gaussian surface shown in figure is



   (1) $72\pi \times 10^3\ Nm^2/C$ (2) $36\pi \times 10^3\ Nm^2/C$

   (3) $48\pi \times 10^3\ Nm^2/C$ (4) $12\pi \times 10^3\ Nm^2/C$

4. An electric dipole is placed at origin, directed along +ve x axis. Consider two points A and B equidistant from dipole as shown in figure. If $\vec{E}_A$ and $\vec{E}_B$ are electric field at A and B, then mark correct option.



   (1) $\vec{E}_A = -\vec{E}_B$ (2) $\vec{E}_A = \vec{E}_B$

   (3) $|\vec{E}_A| > |\vec{E}_B|$ (4) $|\vec{E}_A| < |\vec{E}_B|$

5. Separation between electric dipole of dipole moment p and point charge q kept along axis of dipole is r, then magnitude of force on dipole due to charge is

   (1) 0

   (2) $\frac{kpq}{r^2}$

   (3) $\frac{kpq}{r^3}$

   (4) $\frac{2kpq}{r^3}$

6. Which of the following is incorrect regarding electrostatic field lines?

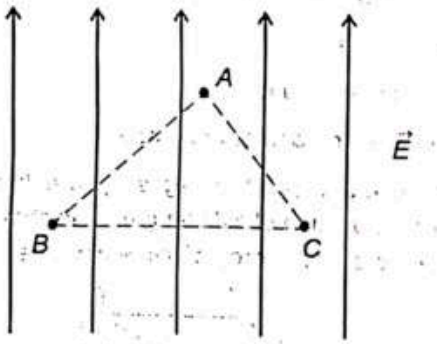
   (1) The tangent to a field line at any point gives the direction of electric field at that point

   (2) Electric field lines always begin from a positive charge and end on a negative charge

   (3) Two field lines can never intersect

   (4) The field lines can form closed loops

7. A uniform electric field having strength $\vec{E}$ exists in a region shown in figure. Consider three points A, B and C having potential V_A, V_B and V_C respectively, then choose the correct option.



(1) $V_A < V_B = V_C$ (2) $V_A > V_B = V_C$

(3) $V_A > V_B > V_C$ (4) $V_B < V_C > V_A$

8. Consider the following two statements and choose the correct option.

Statement-I: Electric field inside conductor is always zero.

Statement-II: Under electrostatic condition electric field inside conductor is non-zero.

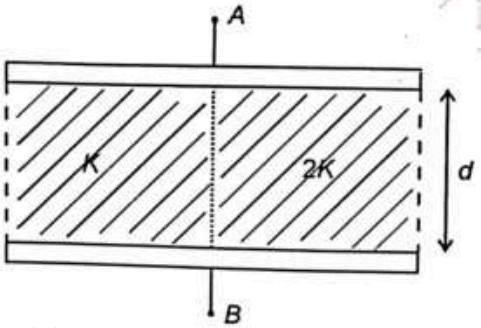
(1) Only statement I is true

(2) Only statement II is true

(3) Both statement I and statement II are true

(4) Neither statement I nor statement II are true

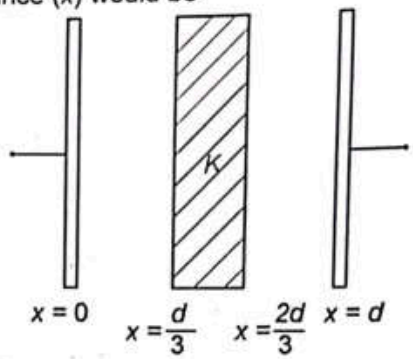
9. Two conducting plates of plate area A and separation 'd' filled with dielectric medium as shown in figure. Given that dimension of both dielectric are same. Equivalent capacitance of the system, will be



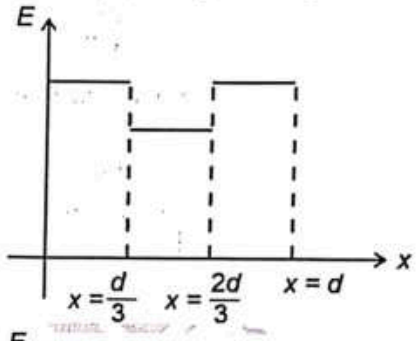
(1) $\frac{K\varepsilon_0 A}{2d}$ (2) $\frac{3K\varepsilon_0 A}{2d}$

(3) $\frac{3K\varepsilon_0 A}{4d}$ (4) $\frac{3K\varepsilon_0 A}{d}$

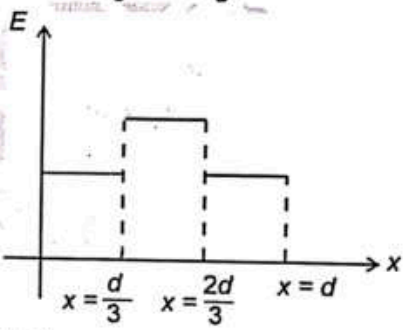
10. A charged capacitor with dielectric medium shown in figure. The correct variation of electric field vs distance (x) would be



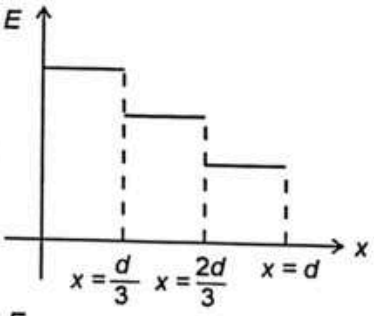
(1)



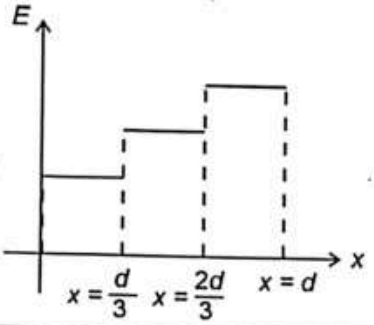
(2)



(3)

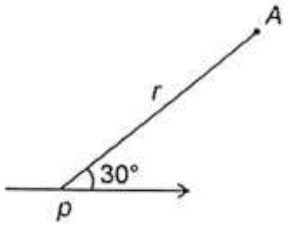


(4)



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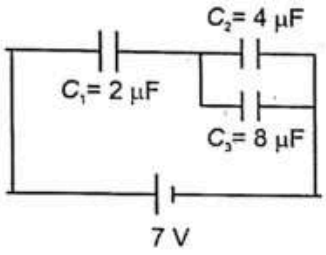
11. Electric potential at A due to dipole 'p' given below is



(1) $\frac{kp}{2r^2}$ (2) $\frac{\sqrt{3}\,kp}{2r^2}$

(3) $\frac{kp}{r^2}$ (4) $\frac{1\,kp}{\sqrt{2}\,r^2}$

12. Charge on capacitor C_1 in the circuit as shown in figure will be

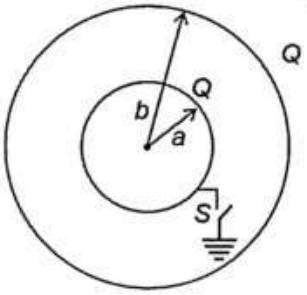


(1) 24 μC (2) 6 μC

(3) 12 μC (4) 8 μC

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13. Initially, both concentric hollow conducting shells of radii a and b have same charge 'Q' shown in figure and the switch is open. Find the flow of charge through the switch when it is closed.



(1) $Q\left(\frac{a+b}{a}\right)$

(2) $Q\left(\frac{a+b}{b}\right)$

(3) $\frac{Qa}{b}$

(4) Q

14. When the separation between two positive point charges is increased, the electrostatic potential energy of the system

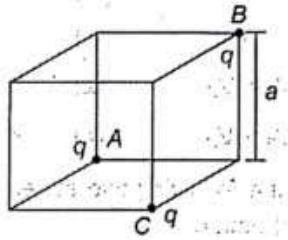
(1) Increases

(2) Decreases

(3) Remains same

(4) May increase or decrease

15. Three equal point charges 'q' are kept at three corners of cube of side length 'a' as shown in figure, then potential energy of system will be



(1) $\frac{kq^2}{a}\left[\frac{1}{\sqrt{2}}+2\right]$

(2) $\frac{kq^2}{a}\left[\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}\right]$

(3) $\frac{kq^2}{a}\left[\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{2}}+\frac{1}{2}\right]$

(4) $\frac{3kq^2}{\sqrt{2}a}$

16. Charge stored on capacitor of capacitance 5 μF when it is charged through battery of emf 5 V is

(1) 1 μC

(2) 25 μC

(3) 25 C

(4) 5 μC

17. A bubble is charged to a potential equal to 6 V. If its radius is tripled, then the potential of the bubble becomes

(1) $\frac{1}{2}$ V

(2) 1 V

(3) 2 V

(4) 18 V

18. An electric dipole having dipole moment $\vec{P}$ is placed in a uniform electric field $\vec{E}$. Then the
    (1) Potential energy is maximum when $\vec{P}$ is perpendicular to $\vec{E}$
    (2) Potential energy is minimum when $\vec{P}$ is perpendicular to $\vec{E}$
    (3) Torque experienced by the dipole is maximum when $\vec{P}$ is parallel to $\vec{E}$
    (4) Torque experienced by the dipole is maximum when $\vec{P}$ is perpendicular to $\vec{E}$
19. In a parallel plate capacitor, the capacitance increases, if
    (1) Distance between the plates increases
    (2) Area of the plate increases
    (3) Area of the plate decreases
    (4) Dielectric constant decreases
20. **Statement I:** Total charge of an isolated system remains conserved.

    Statement II: Charge is a scalar quantity.
    (1) Both statements I and II are true and statement II is correct explanation of statement I
    (2) Both statements I and II are true but statement II is not correct explanation of I
    (3) Statement I is true and statement II is false
    (4) Both Statements I and II are false

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MM : 720

TEST - I

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer :

1. A regular polygon has 10 sides. Equal charges, each Q, are placed at 9 vertices of the polygon and a charge q is placed at the centre of polygon. If the distance of each vertex from its centre is r, then net force experienced by charge q is

   (1) $\frac{Qq}{2\pi\varepsilon_0 r^2}$

   (2) $\frac{Qq}{4\pi\varepsilon_0 r^2}$

   (3) $\frac{9Qq}{4\pi\varepsilon_0 r^2}$

   (4) Zero

2. Three charges $9q$, Q and $4q$ are placed on x-axis, at points $x = 0$, $x = a$ and $x = 2a$ respectively. What should be the value of Q in order to make the net force on $4q$ to be zero?

   (1) $4q$ (2) $-4q$

   (3) $\frac{9q}{4}$ (4) $\frac{-9q}{4}$

Goswami Ji

3. The electric field due to a short dipole at a distance r_1 on the axial line is 16 times as that of electric field at a distance r_2, on the equatorial line. The ratio of $\frac{r_1}{r_2}$ is

   (1) 1 : 1 (2) 1 : 2

   (3) 1 : 16 (4) 1 : 8

4. Given below are two statements, Assertion (A) and Reason (R).

   Assertion (A): Electric field on the surface of an isolated charged conductor is stronger at sharp corners.

   Reason (R) : Surface charge density at any point on the surface of an isolated conductor is inversely proportional to radius of curvature at that point.

   In the light of above statements, select the most appropriate answer from the options given below.

   (1) Both (A) and (R) are true and (R) is the correct explanation of (A)

   (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

   (3) (A) is true but (R) is false

   (4) Both (A) and (R) are false

5. If an electron is brought towards another charge, then electric potential energy of the system

   (1) Decreases

   (2) Increases

   (3) Becomes zero

   (4) May increases or decreases

6. A parallel plate capacitor has an electric field of 10^4 V/m between the plates. If charge on the capacitor is 10 μC, then force on each of the plates is

   (1) 10 mN (2) 100 mN

   (3) 5 mN (4) 50 mN

Space for Rough Work

7. Consider the following statements regarding equipotential surface and choose the correct option.

(A) Equipotential surfaces can never cross each other.

(B) Electric field lines are perpendicular to equipotential surface.

(C) For uniform electric field equipotential surfaces are parallel planes.

(D) The intensity of electric field along an equipotential surface is always zero.

(1) Only A and B are correct

(2) Only A, B and C are correct

(3) Only B, C and D are correct

(4) All A, B, C and D are correct

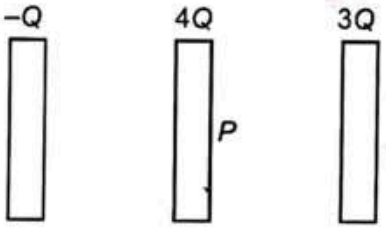
8. An electric field is given by $\vec{E} = (2\hat{i} - \hat{j} + 2\hat{k})$ N/C. The work done in moving a 1 C charge from position $\overrightarrow{r_A} = (\hat{i} + \hat{j})$ m to $\overrightarrow{r_B} = (-\hat{i} + 2\hat{j})$ m against the electric field is

(1) 4 J (2) 2 J

(3) 5 J (4) 3 J

Goswami Ji

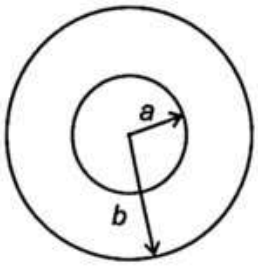
9. Figure shows three large parallel metallic plates with charges $-Q$, $4Q$ and $3Q$ respectively. In steady state, charge on surface P will be



(1) Zero (2) $4Q$

(3) $3Q$ (4) $-3Q$

10. A charged conducting sphere and a charged spherical shell are placed concentric as shown in the figure. If the electric potential of the inner sphere is 13 V and that of the outer shell is 8 V, then the potential at the centre is



(1) 13 V

(2) 8 V

(3) 21 V

(4) 5 V

11. A point charge 100 μC is located in the xy plane at a point having position vector $\overrightarrow{r_0} = (\hat{i} + \hat{j})$ m. The electric field at the point having position vector, $\vec{r} = (2\hat{i} - \hat{j})$ m is

(1) 180 V/m

(2) 180×10^3 V/m

(3) 90×10^4 V/m

(4) 9×10^4 V/m

12. A charged particle having charge +4 C is experiencing an electric force of 100 N towards west at point A. The electric field at point A is

(1) 25 N/C towards east

(2) 25 N/C towards west

(3) 50 N/C towards east

(4) 50 N/C towards west

Space for Rough Work

13. Consider the following statements and choose the correct option.

Statement I: A negative point charge tends to move from a region of low electric potential to a region of high electric potential.

Statement II: The electric potential of a negatively charged conductor must be negative.

Statement III: If $\vec{E}=0$ at a point then V must be zero at that point.

Statement IV: If $V = 0$ at a point then $\vec{E}$ must be zero at that point.

(1) Only statement I is correct

(2) Only statements III and IV are correct

(3) Only statement II is correct

(4) Only statements I and II are correct

14. Two identical small spheres are suspended from the same point by two threads 1 m long. Each sphere is given a charge 20 nC. Finally, they remain in equilibrium at a distance of 20 cm from each other. The mass of each sphere is (take $g = 10\ m/s^2$)

(1) $27\sqrt{11}$ kg (2) $27\sqrt{11}$ ng

(3) $27\sqrt{11}$ g (4) $27\sqrt{11}$ mg

15. An electric dipole of dipole moment $\vec{P}=\left(-4\hat{i}+2\hat{j}-4\hat{k}\right)$ C m is placed in a uniform electric field $\vec{E}=\left(2\hat{i}-\hat{j}+2\hat{k}\right)$ N C^{-1}. The torque (in N m) acting on dipole is

(1) Zero

(2) $4\hat{i}-4\hat{j}+8\hat{k}$

(3) $4\hat{i}+4\hat{j}-8\hat{k}$

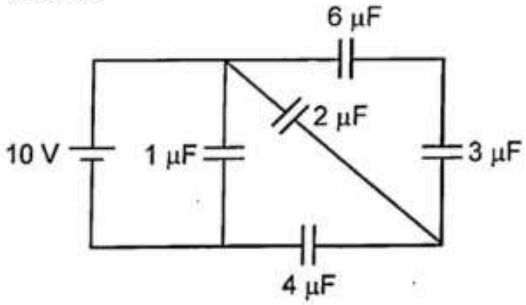
(4) $-4\hat{i}+4\hat{j}+8\hat{k}$

16. If a solid conducting sphere of radius 2 m having charge 2 μC is surrounded by a concentric neutral conducting shell of radius 4 m. If both are connected by a thin conducting wire then the final potential of shell will be

(1) 9 kV (2) 4.5 kV

(3) 2.25 kV (4) 5 kV

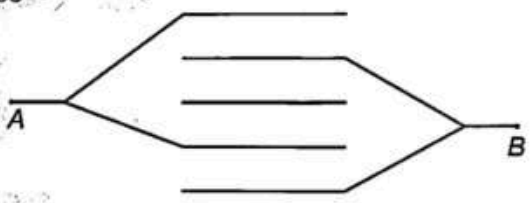
17. The charge stored on 4 μF capacitor in the given circuit is



(1) 10 μC (2) 20 μC

(3) 40 μC (4) 80 μC

18. Five plates each of area A and having separation d between consecutive plates are arranged as shown. Equivalent capacitance between points A and B will be



(1) $\frac{4A\varepsilon_0}{d}$ (2) $\frac{A\varepsilon_0}{4d}$

(3) $\frac{5A\varepsilon_0}{2d}$ (4) $\frac{3A\varepsilon_0}{d}$

19. When 10^{18} electrons are removed from a neutral metal sphere, the charge on the sphere will be

(1) 1.6 C (2) 16 C

(3) 0.16 C (4) 0.016 C

Space for Rough Work

Goswami Ji

20. An electric dipole is situated in an electric field of uniform intensity E whose dipole moment is P and moment of inertia is I. If the dipole is disturbed slightly from the equilibrium position and released, then the time period of its oscillations is

(1) $\sqrt{\frac{I}{PE}}$ (2) $2\pi\sqrt{\frac{I}{PE}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{PE}{I}}$ (4) $\sqrt{\frac{PE}{I}}$

21. If electric potential V at any point (x, y, z) m in space is given by $V = 6x^2$ volt. The electric field at the point $(2, 0, -1)$ m is

(1) $-12\hat{i}$ V/m (2) $-24\hat{i}$ V/m

(3) $-8\hat{i}$ V/m (4) $-16\hat{i}$ V/m

22. Given below are two statements, Assertion (A) and Reason (R).

Assertion (A): For a charged particle moving from one point to another point, the net work done by an electrostatic field on the particle is independent of the path connecting these points.

Goswami Ji

Reason (R) : The net work done by a conservative field depends on end points and is independent of path.

In the light of above statements, select the most appropriate answer from the option given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

23. A conducting hollow sphere having an inner radius r and outer radius $2r$ carries a net charge Q. If a point charge $-q$ is placed at the centre of this sphere, then the surface charge density at inner surface of the sphere will be

(1) $\frac{q+Q}{4\pi r^2}$ (2) $\frac{q+Q}{16\pi r^2}$

(3) $\frac{q}{4\pi r^2}$ (4) $\frac{q}{16\pi r^2}$

24. A simple pendulum consists of a small sphere of mass m suspended by a thread of length L. The sphere carries a positive charge q. The pendulum is made to oscillate in a uniform electric field of strength E directed vertically upward, the time period of oscillations is

(1) $2\pi\sqrt{\frac{L}{g}}$ (2) $2\pi\sqrt{\frac{L}{g+\frac{qE}{m}}}$

(3) $2\pi\sqrt{\frac{L}{g-\frac{qE}{m}}}$ (4) $2\pi\sqrt{\frac{L}{\left(g^2+\left(\frac{qE^2}{m}\right)\right)^{\frac{1}{2}}}}$

25. A charged particle of charge q and mass m moving with a velocity of $u_0\hat{i}$ enters a uniform electric field of strength $\vec{E} = E_0\hat{j}$. The magnitude of displacement of the particle after time t_0 is

(1) $\sqrt{u_0t_0 + \frac{1}{2}\frac{qE_0t_0^2}{m}}$

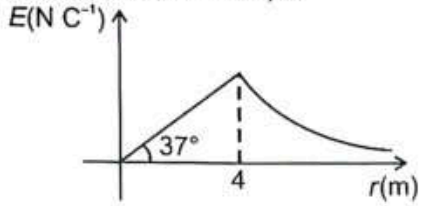
(2) $\sqrt{(u_0t_0)^2 + \left(\frac{1}{2}\frac{qE_0t_0^2}{m}\right)^2}$

(3) $\frac{1}{2}\frac{qE_0t_0^2}{m}$

(4) u_0t_0

Space for Rough Work

26. We have a solid non conducting uniformly charged sphere. Its electric field variation with distance from its centre is shown in graph, then the net charge inside the sphere (in SI unit) is



(1) 85.33 nC (2) 5.33 nC
(3) 144 nC (4) 8.53 μC

27. Two concentric shells kept in air have radii R and $4R$ and surface charge densities σ and $-\sigma$ respectively. The electrical potential at the common centre is

(1) $\frac{3\sigma R}{\varepsilon_0}$ (2) $\frac{-3\sigma R}{\varepsilon_0}$
(3) $\frac{\sigma R}{3\varepsilon_0}$ (4) $\frac{-\sigma R}{3\varepsilon_0}$

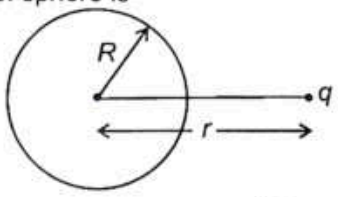
28. Five capacitors of equal capacitance have an equivalent capacitance C_1 when connected in series and an equivalent capacitance C_2 when connected in parallel. The ratio $\frac{C_1}{C_2}$ is

(1) 25 (2) $\frac{1}{25}$
(3) 5 (4) $\frac{1}{5}$

29. Two point charges q and $-2q$ are placed at a distance r. The work done by some external force to increase their separation to $2r$ slowly, will be

(1) $\frac{-Kq^2}{r}$ (2) $\frac{Kq^2}{r}$
(3) $\frac{2Kq^2}{r}$ (4) $\frac{-2Kq^2}{r}$

30. A charge q is kept in front of neutral metallic sphere as shown in the figure. The value of electric potential at the centre of sphere due to induced charge on surface of sphere is

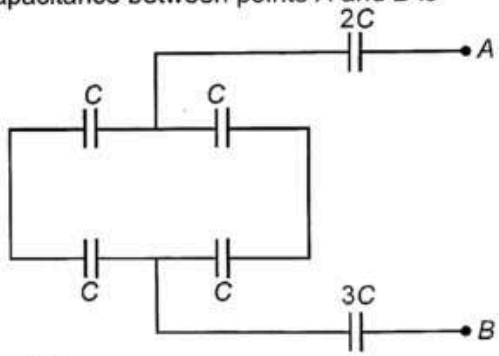


(1) $\frac{Kq}{r}$ (2) $\frac{Kq}{R}$
(3) $\frac{-Kq}{r}$ (4) Zero

31. Two identical capacitors have the same capacitance C. One of them is charged to potential V and other to $2V$. The positive plate of one capacitor is joined with the positive plate of the other capacitor then the loss in energy of combined system in steady will be

(1) CV^2 (2) $\frac{CV^2}{2}$
(3) $2CV^2$ (4) $\frac{CV^2}{4}$

32. In the given arrangement, the value of equivalent capacitance between points A and B is



(1) $\frac{6C}{11}$ (2) $6C$
(3) $\frac{6C}{5}$ (4) $\frac{3C}{2}$

Space for Rough Work

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33. A capacitor is charged by a battery and the energy stored in it is U. The battery is now removed and the separation between the plates is halved. Now the energy stored is

(1) U (2) $2U$

(3) $\frac{U}{2}$ (4) $4U$

34. Two symmetrical large parallel metal plates having charges Q and $2Q$ are placed at small separation. Considering area of one side of a plate to be A, what will be the electric field at a point in between the plates?

(1) Zero (2) $\frac{Q}{2A\varepsilon_0}$

(3) $\frac{Q}{A\varepsilon_0}$ (4) $\frac{2Q}{A\varepsilon_0}$

35. Consider the following statements and choose the correct option.

Statement I: Value of charge increases with increase in its speed.

Statement II: Charge on any object is quantised.

Statement III: For an isolated system net charge is conserved.

Statement IV: Specific charge on any object is invariant.

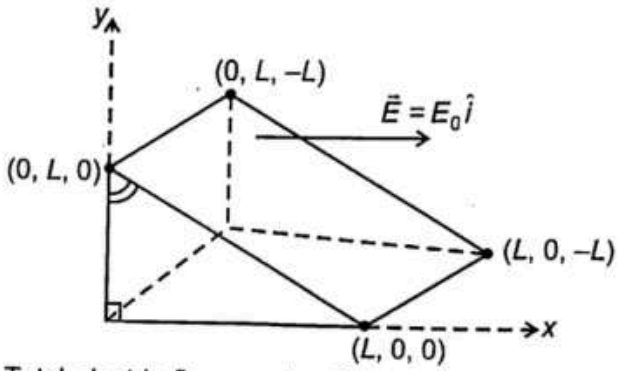
(1) Only statement II is correct

(2) Only statements II and III are correct

(3) Only statements I, II and III are correct

(4) Only statements II, III and IV are correct

36. A horizontal oriented, uniform electric field of magnitude E_0 passes through a triangular prism as shown in figure below.



Total electric flux passing through slant surface of the prism will be

(1) E_0L^2 (2) $\frac{E_0L^2}{\sqrt{2}}$

(3) Zero (4) $\frac{E_0L^2}{2}$

37. A charge $4Q$ is distributed uniformly along the periphery of a semicircular ring of radius R. The linear charge density of the ring is

(1) $\frac{Q}{4\pi}$ (2) $\frac{Q}{2\pi}$

(3) $\frac{2Q}{\pi R}$ (4) $\frac{4Q}{\pi R}$

38. A charged particle of mass m and charge q is released in an uniform electric field of magnitude E. Its speed after time t will be

(1) qEt (2) $\frac{qEt^2}{m}$

(3) $\frac{mqt}{E}$ (4) $\frac{qE}{m}t$

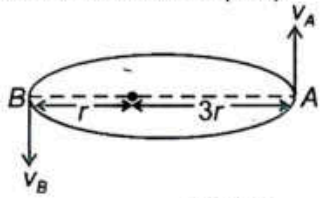
39. Three metallic spheres say A, B and C have charges 17 C, –13 C and 4 C respectively. A, B and C are brought in contact such that final charge on each sphere A and B becomes 6 C, the charge on sphere C is

(1) 4 C (2) –4 C

(3) 6 C (4) 2 C

Space for Rough Work

40. A positive charged particle is revolving around a fixed negative charged particle in an elliptical orbit. Minimum and maximum distance of positive charged particle from negative charged particle are r and $3r$ respectively. If maximum kinetic energy of positive charged particle is 9 J. The work done by electric force during motion of positive charged particles from A to B will be (in J)



(1) 3 J (2) 9 J
(3) 8 J (4) 7 J

41. A point charge q is placed at the centre of an edge of a cube of side length l. The electric flux through the cube is

(1) $\frac{q}{8\varepsilon_0}$ (2) $\frac{q}{4\varepsilon_0}$
(3) $\frac{q}{2\varepsilon_0}$ (4) $\frac{q}{\varepsilon_0}$

42. An electric dipole with dipole moment 10^{-5} C m is aligned with the direction of a uniform electric field of magnitude 2×10^5 N/C. The dipole is then rotated through an angle of 37° with respect to the electric field. The change in the potential energy of the dipole is

(1) 2 J (2) 0.2 J
(3) 0.4 J (4) 4 J

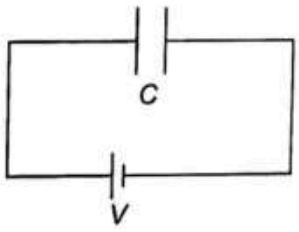
43. A point charge q is placed very close to the centre of a circular surface of radius R. The flux crossing the circular surface would be nearly

(1) $\frac{q}{2\varepsilon_0}$ (2) $\frac{q}{\varepsilon_0}$
(3) $\frac{q}{\pi\varepsilon_0}$ (4) $\frac{q}{2\pi R\varepsilon_0}$

44. There are 64 drops of a conducting fluid. Each has radius r and they are charged to a potential V. They are then combined to form a bigger drop. The potential of bigger drop will be

(1) V
(2) $4V$
(3) $16V$
(4) $64V$

45.



An air filled parallel plate capacitor having capacitance, C is connected with a battery of voltage V. It has charge Q, electric field E and energy density u. Now, battery is disconnected from capacitor and a dielectric plate is fully inserted between plates. Match the following columns on the basis of above information.

	Column-I (Quantities)		Column-II (Their Values (finally))
A.	Charge (Q)	(P)	Decreases
B.	Capacitance (C)	(Q)	Increases
C.	Electric field (E)	(R)	Remain same
D.	Energy density (u)	(S)	Becomes zero

(1) A-R, B-Q, C-S, D-S
(2) A-R, B-Q, C-P, D-P
(3) A-S, B-Q, C-R, D-R
(4) A-S, B-Q, C-P, D-P

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[CHEMISTRY]

46. Mass of 90% pure $CaCO_3$ required to neutralise 100 mL 0.4 M HCl solution will be

(1) 1.11 g (2) 2.22 g

(3) 3.33 g (4) 4.44 g

47. Boiling point of NaCl (aq) solution in which mole fraction of water is $\frac{9}{10}$ will be (K_b of H_2O = 0.52 K kg mol^{-1} and NaCl supposed to be completely ionised)

(1) 101.4 °C

(2) 120.66 °C

(3) 106.4 °C

(4) 100.52 °C

48. Consider the following statements

(a) For a solution of HNO_3 (aq), boiling point of azeotropic solution will be less than both pure HNO_3 and H_2O.

(b) An ideal solution cannot form azeotropic mixture.

(c) Reverse osmosis is a spontaneous process.

(d) On addition of small amount of HgI_2 in aqueous solution of KI, boiling point of resultant solution increases.

(e) When two isotonic solutions are separated by semipermeable membrane no osmosis occurs between them.

Correct statements are

(1) (a), (d) and (e) only

(2) (b) and (d) only

(3) (a), (c), (d) and (e) only

(4) (b) and (e) only

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49. Match List-I with List-II.

	List-I		List-II van't Hoff factor (i)
a.	Na_2CO_3 (Degree of dissociation 70%)	(i)	3.4
b.	$K_4[Fe(CN)_6]$ (Degree of dissociation 60%)	(ii)	1.8
c.	$Al_2(SO_4)_3$ (Degree of dissociation 75%)	(iii)	2.4
d.	C_6H_5COOH (Degree of dissociation 80%)	(iv)	4

Correct match is

(1) a(i), b(iii), c(iv), d(ii) (2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(i), c(ii), d(iv) (4) a(iii), b(i), c(iv), d(ii)

50. Given below are two statements.

Statement I: If a solution is prepared by adding 1 g non-volatile non-electrolyte solute A in 1000 g water gives higher vapour pressure than a solution of 1 g $C_6H_{12}O_6$ in 1000 g water then molar mass of A will be more than 180 g mol^{-1}.

Statement II: If a binary solution is obtained by mixing equal number of moles of two volatile liquids. At equilibrium, vapour phase will be always rich in the component which is more volatile.

In the light of above statements, choose the **most appropriate** answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

Space for Rough Work

51. Given below are two statements.

Statement I: 0.1 m NaCl (aq) has higher boiling point than 0.1 m KCl (aq) if both are completely ionised.

Statement II: Molal elevation constant for (aq) NaCl solution is higher than (aq) KCl solution.

In the light of above statements, choose the **most appropriate** options from the options given below.

(1) Both statement I and statement II are incorrect
(2) Statement I is correct but statement II is incorrect
(3) Both statement I and statement II are correct
(4) Statement I is incorrect but statement II is correct

52. 0.1 M $CaCl_2$ with 80% ionisation will be isotonic with
(1) 0.13 M (aq) KCl with 100% ionisation
(2) 0.3 M (aq) urea
(3) 0.1 M Na_2SO_4 with 90% ionisation
(4) 0.2 M (aq) $C_{12}H_{22}O_{11}$

53. Mass percent of solute in 0.1 m aqueous glucose solution will be
(1) 2.22% (2) 1.76%
(3) 1.05% (4) 2.86%

54. Consider the following statements.
(a) Mole fraction is temperature independent.
(b) A mixture of chloroform and acetone form a solution with negative deviation from Raoult's law.
(c) Solubility of gases in liquids decreases with rise in temperature.
(d) A solution in which no more solute can be dissolved at the fixed temperature and pressure is called saturated solution.

Correct statements are
(1) (a), (b) and (c) only
(2) (a), (b) and (d) only
(3) (a), (b), (c) and (d)
(4) (c) and (d) only

55. Correct order of solubility of following gases in water at 298 K will be
(1) HCHO < CO_2 < CH_4 < Ar
(2) Ar < CO_2 < CH_4 < HCHO
(3) Ar < CH_4 < CO_2 < HCHO
(4) Ar < HCHO < CO_2 < CH_4

56. A mixture of benzene and toluene has 39.52 mm Hg vapour pressure at 300 K. The vapour pressure of pure benzene is 50.71 mm Hg. If mole fraction of toluene in solution is 0.6 then its vapour pressure in pure state will be
(1) 35.06 mm Hg (2) 38.06 mm Hg
(3) 20.08 mm Hg (4) 32.06 mm Hg

57. Mass of ethylene glycol that should be added to 4 kg of water to prevent it from freezing at –6°C will be (K_f = 1.86 K kg mol^{-1})
(1) 204 g (2) 304 g
(3) 400 g (4) 800 g

58. Among 0.1 m solutions of urea, Na_3PO_4 and $Al_2(SO_4)_3$:-
(Consider 100% dissociation of electrolytes)
(a) The vapour pressure and freezing point are the lowest for urea solution
(b) The vapour pressure and freezing point are the highest for urea solution
(c) The elevation in boiling point is the highest for $Al_2(SO_4)_3$
(d) The depression in freezing point is the highest for $Al_2(SO_4)_3$

The correct statement(s) is/are
(1) Only (a) (2) (b) and (c) only
(3) (b), (c) and (d) only (4) (a), (b), (c) and (d)

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Space for Rough Work

59. Given below are two statements.

Statement I: Cooking time is reduced in pressure cooker.

Statement II: Boiling point of H_2O reduces in pressure cooker.

In the light of above statements, choose the **most appropriate** answer from the options given below.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are correct

(4) Statement I is incorrect but statement II is correct

60. Consider the following expressions for aqueous solution of non-volatile and non-electrolyte solute B.

(a) $\frac{P^0 - P_s}{P_s} = \frac{n_B}{n_{H_2O}}$ (b) $P_s \propto X_{H_2O}$

(c) $\frac{P^0 - P_s}{P^0} = X_B$ (d) $\frac{P_s}{P^0} = \frac{X_B}{X_{H_2O}}$

P^0 = Vapour pressure of pure water

P_s = Vapour pressure of solution

X_{H_2O} = Mole fraction of H_2O

X_B = Mole fraction of B

n_B = Number of moles of B

n_{H_2O} = Number of moles of H_2O

Correct expressions are

(1) (a), (b) and (c) only

(2) (b) and (c) only

(3) (c) and (d) only

(4) (a) and (d) only

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61. Cell potential for the given galvanic cell will be

$Zn(s) + 2H^+(aq)(pH = 4) \longrightarrow$

$Zn^{2+}(aq)(0.01\,M) + H_2(g)(2\,atm)$

$\left(E^\circ_{Zn^{2+}/Zn} = -0.76\ V\right)$

(1) 0.84 V

(2) 0.62 V

(3) 0.58 V

(4) 0.92 V

62. Consider the following statements

(a) Ag is more reactive metal than Cu.

(b) Al^{3+}/Al has more negative standard reduction potential than Zn^{2+}/Zn.

(c) $Cl_2 + 2Br^-(aq) \rightarrow Br_2 + 2Cl^-(aq)$ is a spontaneous reaction.

Correct statement(s) is/are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) only

(4) (a), (b) and (c)

63. K_C for given cell representation will be

$Cu(s)\,|\,Cu^{2+}(aq)(0.01\,M)\,||\,Cu^{2+}(aq)(0.1\,M)\,|\,Cu(s)$

(1) 10^2 (2) 10^{-2}

(3) 10 (4) 1

64. Correct reaction of Daniell cell is

(1) $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

(2) $2Al(s) + 3Zn^{2+}(aq) \rightarrow 2Al^{3+}(aq) + 3Zn(s)$

(3) $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

(4) $3Mg(s) + 2Fe^{3+}(aq) \rightarrow 3Mg^{2+}(aq) + 2Fe(s)$

Space for Rough Work

65. Given below are two statements.

Statement I: Galvanic cell converts chemical energy of a spontaneous redox reaction into electrical energy.

Statement II: Electrode potential is the potential difference developed between the electrode and the electrolyte.

In the light of above statements, choose the **correct** option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

66. Given below are two statements.

Statement I: In a galvanic cell, the half cell in which oxidation takes place is anode and it has positive potential with respect to solution.

Statement II: In a galvanic cell, the half cell in which reduction takes place is cathode and it has negative potential with respect to solution.

In the light of above statements, choose the **correct** option.

Goswami Ji

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

67. Given below are two statements, one is labelled as assertion (A) and the other is labelled as reason (R).

Assertion (A): Cu does not dissolve in dilute HCl.

Reason (R) : Standard reduction potential of Cu^{2+} is positive.

In the light of above statements, select the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

68. If standard reduction potential of redox couple (M^{2+}/M) is E then reduction potential if concentration of $[M^{2+}]$ is 0.01 M is

(1) $E - \frac{RT}{F} \ln 10$ (2) $E + \frac{RT}{F} \ln 10$

(3) $E - \frac{RT}{2F} \ln 10$ (4) $E - \frac{RT}{F} \log 10$

69. Consider the following statements

(a) Mg is better reducing agent than Ca.

(b) F_2 is better oxidising agent than Cl_2 in aqueous medium.

(c) Li^+ has minimum standard reduction potential value.

Correct statement(s) is/are

(1) (a) and (c) only (2) (b) and (c) only

(3) (a) and (b) only (4) (c) only

70. Given that the standard potential (E°) of MnO_4^- / MnO_2, MnO_2/Mn^{2+} and Mn^{2+}/Mn are x V, y V and z V respectively, the E° of MnO_4^- / Mn is:

(1) $\frac{x + y + z}{7}$

(2) $x + y + z$

(3) $\frac{3x + 2y + 2z}{7}$

(4) $3x + 2y - 2z$

Space for Rough Work

71. Some standard reduction potential at 298 K are given below.

Sn^{2+}/Sn	–.014 V
Ni^{2+}/Ni	–0.25 V
Fe^{2+}/Fe	–0.44 V
Zn^{2+}/Zn	–0.76 V

To a solution containing 0.002 M of X^{2+} and 0.2 M of Y^{2+}, the metal rods X and Y are inserted (at 298 K) and connected by a conducting wire. This resulted in dissolution of X. The incorrect combination of X and Y, respectively is

(1) Ni and Sn (2) Fe and Ni
(3) Ni and Zn (4) Zn and Sn

72. If standard cell potential for a galvanic cell is 1.41 V then standard Gibbs energy for cell reaction will be

$A^{2+}(aq) + B(s) \rightleftharpoons B^{2+}(aq) + A(s)$

(1) +272.13 kJ/mol (2) –136.06 kJ/mol
(3) +136.06 kJ/mol (4) –272.13 kJ/mol

73. Which among the following material has highest conductivity at 298.15 K?

(1) Fe (2) Cu
(3) Ag (4) Graphite

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74. Given below are two statements.

Statement I: Molar conductivity increases with decrease in concentration of electrolyte.

Statement II: Specific conductance decreases with decrease in concentration of electrolyte.

In the light of above statements, choose the **correct** option.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

75. If conductivity of cell with 200 Ω resistance is 5×10^{-3} ohm^{-1} cm^{-1} then cell constant will be

(1) 0.1 cm^{-1} (2) 1 cm^{-1}
(3) 0.01 cm^{-1} (4) 2.5×10^{-2} cm^{-1}

76. Which among the following ion will have highest limiting molar conductivity in aqueous medium at 298 K?

(1) Cl^- (2) Na^+
(3) Mg^{2+} (4) OH^-

77. If conductivity of 10^{-3} M CH_3COOH solution is 1.85×10^{-5} S cm^{-1} and limiting molar conductivity of H^+ and CH_3COO^- ion are 349.6 S cm^2 mol^{-1} and 40.9 S cm^2 mol^{-1} respectively. The degree of dissociation for 10^{-3} M CH_3COOH will be

(1) 4.7 % (2) 2.3 %
(3) 11.2 % (4) 9.1 %

78. To 500 cm^3 of water, 6 g of acetic acid is added. If 46% of acetic acid is dissociated, what will be the depression in freezing point? (K_f and density of water are 1.86 K kg mol^{-1} and 0.997 g cm^{-3}, respectively)

(1) 0.11°C (2) 0.86°C
(3) 0.54°C (4) 1.08°C

79. Match electrolyte in list-I with amount of charge required to deposit 1 mol metal in list-II.

	List-I		List-II
a.	Aq. $CuSO_4$	(i)	2 F
b.	Molten $AlCl_3$	(ii)	1 F
c.	Molten AgBr	(iii)	$\frac{8}{3}$ F
d.	Motlen Fe_3O_4	(iv)	3 F

Correct match is

(1) a(i), b(iv), c(ii), d(iii) (2) a(ii), b(iv), c(i), d(iii)
(3) a(ii), b(iii), c(i), d(iv) (4) a(i), b(iii), c(ii), d(iv)

Space for Rough Work

80. A solution of (aq) $AgNO_3$ is electrolysed for 10 minutes with a current of 3 A then mass of Ag deposited at cathode will be (Molar mass of Ag = 108 g mol^{-1})

(1) 4.02 g (2) 1.005 g

(3) 2.01 g (4) 8.04 g

81. Given below are two statements, one is labelled as assertion (A) and the other is labelled as reason (R).

Assertion (A): On electrolysis of aqueous NaCl solution using inert electrodes, pH of the solution increases.

Reason (R) : On electrolysis of aqueous NaCl, water gets reduced at cathode and concentration of OH^- ion increases in solution.

In the light of above statements, select the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

82. Match batteries in list-I with their electrolyte in list-II.

	List-I		List-II
a.	Dry cell	(i)	38% (aq) H_2SO_4
b.	Mercury cell	(ii)	Moist paste of NH_4Cl and $ZnCl_2$
c.	($H_2 - O_2$) fuel cell	(iii)	Paste of KOH and ZnO
d.	Lead storage battery	(iv)	Conc. (aq) NaOH solution

Correct match is

(1) a(ii), b(iii), c(i), d(iv)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(ii), b(iii), c(iv), d(i)

83. Which among the following is best metal that can be used as sacrificial electrode, to prevent corrosion of iron?

(1) Sn

(2) Ag

(3) Cu

(4) Mg

84. If 2 A current is passed through 1 L brine solution for 965 seconds then total moles of gas released is

(1) 0.01

(2) 0.03

(3) 0.02

(4) 0.04

85. Standard reduction potential of three metals are in order of $E^\circ_{A^{2+}/A} > E^\circ_{B^{2+}/B} > E^\circ_{C^{2+}/C}$

Identify the correct statement among the following.

(1) A is soluble in BSO_4

(2) C is soluble in $A(NO_3)_2$

(3) C^{2+}(aq) is strongest oxidising agent

(4) B(s) + $CSO_4 \rightarrow BSO_4$ + C(s) will have positive value of EMF

86. Amount of ice separated if a solution of 1000 g water and 0.1 mol glucose is cooled down to −1°C will be (K_f of H_2O = 1.86 K kg mol^{-1})

(1) 200 g

(2) 186 g

(3) 600 g

(4) 814 g

Space for Rough Work

87. When 2.175 g of non-volatile non-electrolyte solute is added in 39 g of benzene, the vapour pressure of solution recorded is 600 mm Hg. If vapour pressure of pure benzene at same temperature as of solution is 640 mm Hg then molar mass of solute will be

(1) 32.75 $gmol^{-1}$ (2) 130.5 $gmol^{-1}$

(3) 90.25 $gmol^{-1}$ (4) 65.25 $gmol^{-1}$

88. Total volume of gas evolved at STP on electrolysis of $CuSO_4$ (aq) using pt electrode by passing 1 F charge will be

(1) 16.8 L (2) 5.6 L

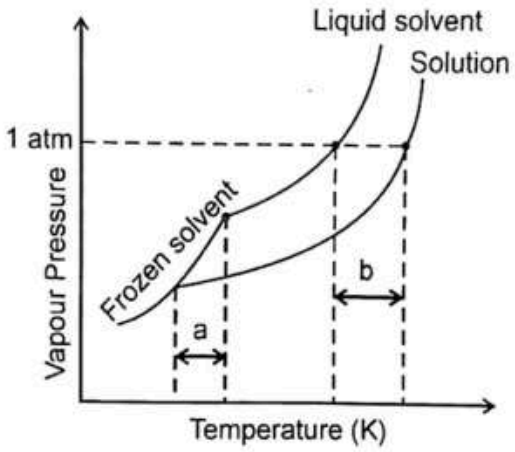
(3) 33.6 L (4) 44.8 L

89. Blood cell fluid is isotonic with

(1) 0.9 m NaCl (2) 0.9%(w/V) NaCl

(3) 0.11 m NaCl (4) 0.11%(w/V) KCl

90.



On the basis of above diagram for the aqueous solution of a non volatile solute, 'a' and 'b' respectively are

(1) ΔT_b and ΔT_f (2) ΔT_f and ΔT_b

(3) K_b and K_f (4) K_f and K_b

[BOTANY]

91. The proximal end of the filament of stamen can be attached to the _____ of the flower.

Goswami Ji

Select the **correct** one(s) to fill the above blank.

(a) Thalamus (b) Tepal

(c) Petal (d) Anther

(1) a and d only

(2) b and c only

(3) a and c only

(4) a, b and c only

92. The ploidy of each cell of a microspore tetrad is

(1) Tetraploid (2) Haploid

(3) Diploid (4) Polyploid

93. Read the below given statements and select the **correct** option.

Statement A: When the pollen grain is mature it contains two cells.

Statement B: In less than 40% of angiosperms, the generative cell of pollen grain divides to give rise to two male gametes after shedding.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

94. The walls of pollen grains do not have

(1) Sporopollenin (2) Pectin

(3) Cellulose (4) Chitin

Space for Rough Work

95. State the given statements as True **(T)** or False **(F)** and select the **correct** option.

A. In an angiospermic embryo sac, only six of the eight nuclei are surrounded by cell walls.

B. The egg apparatus is a seven celled structure in angiosperms.

C. In an ovule generally one megaspore remains functional and forms an embryo sac.

D. Ovules generally differentiate a single megaspore mother cell in the chalazal end.

E. Embryo sac is located in the nucellus.

	A	B	C	D	E
(1)	T	F	T	T	F
(2)	F	F	T	F	T
(3)	T	T	F	T	T
(4)	F	T	T	F	F

96. Pollen grains of which of the following families lose viability within 30 minutes of their release?

(1) Rosaceae (2) Leguminosae
(3) Solanaceae (4) Poaceae

97. In all of the following plants both cleistogamy and chasmogamy are possible, **except**

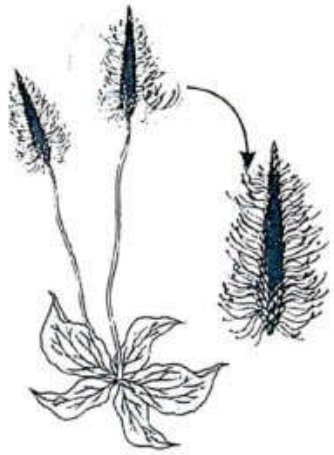
(1) *Viola* (2) *Oxalis*
(3) *Commelina* (4) Maize

98. Mark the statements that are **true** for pollination in majority of flowering plants and choose the correct option.

(a) They use biotic agents
(b) They produce enormous amount of pollen as compared to number of ovules
(c) They offer floral rewards
(d) They produce mucilage covered unwettable pollen grains

(1) (a), (c) and (d) (2) (b), (c) and (d)
(3) (a) and (c) only (4) (a), (b) and (c)

99. Identify the type of pollination that is more likely possible in the given plant. Select the **correct** statements related to it.



(a) It has compact inflorescence
(b) It produces light weight pollen grains
(c) The flowers are always chasmogamous
(d) They produce nectar
(e) The flowers often have large feathery stigma

(1) (a), (b) and (e) only
(2) (a), (b), (c) and (e) only
(3) (b), (c), (d) and (e) only
(4) All (a), (b), (c), (d) and (e)

100. Both autogamy and geitonogamy are prevented in

(1) *Hibiscus* (2) Papaya
(3) Maize (4) Lily

101. How many of the given plants or plant groups show water pollination?

(a) Algae (b) *Vallisneria*
(c) Bryophytes (d) *Hydrilla*
(e) Pteridophytes (f) *Zostera*

(1) 3 (2) 4
(3) 6 (4) 5

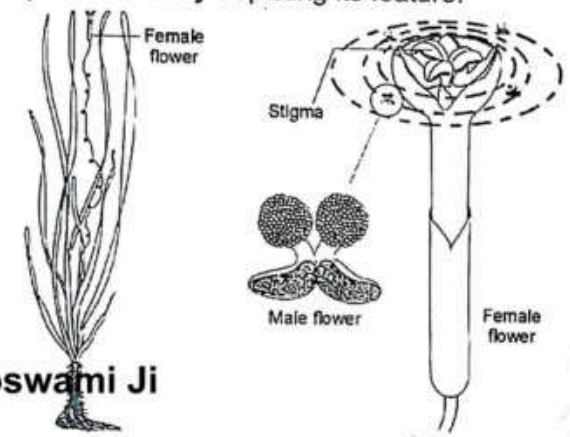
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Goswami Ji

102. In majority of aquatic plants, flowers are pollinated by
(1) Water (2) Insects or wind
(3) Birds (4) Butterflies

103. Mark the statement which is **not** true for pollination.
(1) Among the animals, bees are the dominant pollinating agents.
(2) The large sized animals do not act as pollinators.
(3) Flowers pollinated by beetles produce foul odour.
(4) Floral rewards are must in animal pollinated flowers.

104. Identify the plant in below given figure and mark the option **correctly** depicting its feature.



(a) It is a fresh water plant popularly called sea grass.
(b) It shows epihydrophily.
(c) It releases its male flowers or pollen grains on the surface of water.
(d) Female flowers are produced on the long stalk.
(e) It produces long ribbon like pollen.
(1) (a), (b), (c) and (e) are true
(2) (a), (c) and (d) are true
(3) (b), (d) and (e) are true
(4) (b), (c) and (d) are true

105. All of the following plants may get pollinated by insects, **except**
(1) *Yucca* (2) *Amorphophallus*
(3) Water lily (4) Rice

106. The structure that guides the entry of pollen tube into the embryo sac is
(1) Filiform apparatus (2) Synergid
(3) Antipodal (4) Polar nuclei

107. Identify the **wrongly** matched pair.

(1)	Castor	–	Unisexual flowers
(2)	Self pollination	–	Inbreeding depression
(3)	*Papaver*	–	Multicarpellary apocarpous ovary
(4)	Anatropous ovule	–	A typical angiospermic ovule

108. All of the following are outbreeding devices except the one. Mark the **odd** one among these.
(1) Self incompatibility
(2) Dioecy
(3) Hermaphrodite flower
(4) Placement of stigma and anther at different positions

109. Which of the given statements is/are **true** for castor?
(a) It prevents both autogamy and geitonogamy
(b) It has albuminous seeds
(c) It is a dioecious plant
(d) The developing embryo inside the seed consumes the endosperm completely
(1) (a), (b) and (d) (2) (c) and (d)
(3) (a) and (b) only (4) (b) only

Goswami Ji

Space for Rough Work

110. Which of the given steps of artificial hybridisation is not required if a breeder is interested in crossing maize plants?
(1) Bagging
(2) Emasculation of parent flower
(3) Rebagging
(4) Collection of pollen

111. Mark the conversion which does not occur after double fertilisation in an angiospermic plant.
(1) Central cell → Primary endosperm cell
(2) Zygote → Embryo
(3) Fusion of polar nuclei → Secondary nucleus
(4) Primary endosperm nucleus → Endosperm

112. Read the following Assertion **(A)** and Reason **(R)** and select the **correct** option.
Assertion (A): In some plant species, diploid egg cell is formed without reduction division.
Reason (R): In some plant species, egg cell directly develops into embryo without fertilisation.
(1) Assertion is true but reason is false statement
(2) Assertion is false but reason is true statement
(3) Both assertion and reason are true and reason is the correct explanation of assertion
(4) Both assertion and reason are true, but reason is not the correct explanation of assertion

Goswami Ji

113. Which one is **not** true for a typical anther?
(1) Bilobed (2) Dithecous
(3) Trigonal (4) Tetrasporangiate

114. Perisperm is
(1) Found only in endospermic monocot seeds
(2) Residual endosperm
(3) Diploid tissue
(4) Persistent integument of ovule

115. Mark the **odd** one w.r.t. false fruits.
(1) Cashew (2) Strawberry
(3) Apple (4) Mango

116. Which of the given anther wall layer does not help in its dehiscence to release the pollen grains?
(1) Epidermis (2) Middle layer
(3) Endothecium (4) Tapetum

117. Pollen grains
(1) Measure about 25-50 mm in diameter
(2) Are well preserved as fossils because of intine layer
(3) May cause chronic respiratory disorders
(4) Represents female gametophyte

118. Read the below given statements for flowering plants and select the **correct** option.
A. Mature male gametophytes undergo two mitotic divisions to achieve three celled stage.
B. Pollen grains can be stored at –96°C for years in liquid nitrogen.
C. Orchids have more number of ovules in an ovary as compared to paddy.
D. In an ovule, integuments and nucellus are absent at the micropylar end.
(1) Only A, B and D are true
(2) Only C is true
(3) Only A, C and D are true
(4) All A, B, C and D are true

119. Select the **wrongly** matched pair.

(1)	Zygote	–	Diploid
(2)	Endosperm	–	Triploid
(3)	Megaspore mother cell	–	Diploid
(4)	Scutellum	–	Haploid

Space for Rough Work

120. Both water pollinated and wind pollinated plants
(1) Produce large sized flowers
(2) Form nectarless flowers
(3) Have large feathery stigma
(4) Have colourful flowers

121. Read the statement I and II and mark the **correct** option.
Statement I: Wind pollination is quite common in grasses and it is absent in aquatic plants.
Statement II: The most common type of embryo sac is monosporic with only two sequential mitotic divisions.
(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statements I and II are correct
(4) Both statements I and II are incorrect

122. Double fertilisation is seen in
(1) Flowering plants
(2) All seed producing plants
(3) All green plants
(4) All those plants that produce pollen

123. Coconut water from tender coconut is
(1) Endosperm (2) Nucellus
(3) Perisperm (4) Embryo

124. In monocot embryo, root cap is enclosed in undifferentiated sheath like structure called
(1) Coleoptile (2) Coleorhiza
(3) Plumule (4) Radicle

125. Each seed of orange contains
(1) Many embryos of different shapes
(2) Both haploid and diploid embryos
(3) Only one embryo
(4) Only haploid embryos

126. In mango and citrus, embryo may arise from
(a) Zygote
(b) Nucellus
(c) Endosperm
Mark the **correct** one(s).
(1) a only (2) a and c only
(3) b and c only (4) a and b only

127. During megasporogenesis in flowering plants, archesporium differentiates from
(1) Nucellus (2) Chalaza
(3) Integuments (4) Megaspore

128. Parasitic plant containing many seeds in a fruit is
(1) *Striga* (2) Papaya
(3) Watermelon (4) Wheat

129. All of the following form apomictic embryos, **except**
(1) Diploid egg cell (2) Nucellus
(3) Integument (4) Zygote

130. Xenogamy leads to
(1) Fertilisation of a flower by the pollen from another flower of same plant
(2) Fertilisation of a flower by the pollen from a flower of another plant of same population
(3) Fertilisation of a flower by the pollen from the same flower
(4) Fertilisation of a flower by the pollen from a flower of another plant belonging to distinct population

131. Which one is **not** a correct statement?
(1) False fruits arise from other floral parts except ovary
(2) True fruits are not found in strawberry
(3) Fruits that develop without fertilisation are called parthenocarpic
(4) Application of growth hormones can induce parthenocarpy

Space for Rough Work

132. Germ pore
(1) Facilitates emergence of radicle
(2) Absorbs water for seed germination
(3) Allows growth of pollen tube
(4) Attaches pollen to anther wall

133. Mark the parthenocarpic fruit among these.
(1) Strawberry (2) Banana
(3) Cashewnut (4) Mango

134. The oldest viable seed is of
(1) Date palm (2) *Phoenix*
(3) Orchid (4) Lupine

135. Pollen-pistil interaction does **not** involve
(1) Pollen deposition on stigma
(2) Pollen tube growth and entry in ovule
(3) Pollen germination
(4) Pollen release from anther and transfer to stigma of a flower

[ZOOLOGY]

136. Complete the analogy by selecting the **correct** option w.r.t. external genitalia of humans.
Penis : Male external genitalia :: _____ : Female external genitalia
(1) Vagina (2) Cervix
(3) Clitoris (4) Womb

137. The proteinaceous hormone secreted by ovary only during pregnancy is
(1) Estrogen
(2) hPL
(3) Progesterone
(4) Relaxin

138. Choose the mismatch.

(1)	Cells of corona radiata	–	Ploidy level is same as that of primary oocyte
(2)	Zona pellucida	–	Cellular layer secreted by secondary follicle
(3)	Perivitelline space	–	Lies between zona pellucida and cell membrane of an ootid
(4)	First polar body	–	Formed prior to ovulation

139. If an adult male human ejaculates 200-300 million sperms during a coitus, then at least, how many sperms out of these must have normal shape and size as well as vigorous motility for normal fertility?
(1) 48-72 million (2) 78-96 million
(3) 180-240 million (4) 100-200 million

140. Consider the following
(a) Partially covers the opening of vagina
(b) Presence or absence is not a reliable indicator of virginity
(c) Membranous structure included under female external genitalia
Select the option for which all of the above features hold true.
(1) Labia minora (2) Labia majora
(3) Hymen (4) Mons pubis

141. In human females, meiosis-II of secondary oocyte is completed during _____ in _____.
Select the option that correctly fills the respective blanks.
(1) Implantation ; Uterus
(2) Fertilisation ; Fallopian tube
(3) Ovulation ; Oviduct
(4) Menstruation ; Womb

Space for Rough Work

142. Match column I with column II and select the correct option w.r.t. menstrual cycle in human females.

	Column I		Column II
a.	Follicular phase	(i)	High levels of progesterone
b.	Luteal phase	(ii)	Gradual increase in gonadotropins
c.	Ovulatory phase	(iii)	Low levels of ovarian hormones
d.	Menstrual phase	(iv)	Peak levels of FSH as well as LH

(1) a(i), b(ii), c(iii), d(iv)
(2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(i), c(ii), d(iv)
(4) a(iv), b(iii), c(ii), d(i)

143. In case of humans, spermatozoa can be differentiated from ovum in all of the given basis, **except**
(1) Presence of locomotory structure
(2) Amount of cytoplasm
(3) Chromosome number
(4) Presence of spirally arranged mitochondria

Goswami Ji

144. Breast feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby as colostrum majorly contains high amount of 'X' absolutely essential to develop resistance for the new born babies. Identify 'X' and choose the correct option.
(1) Antigens
(2) Antibodies
(3) Hormones
(4) Enzymes

145. How many of the structures mentioned in the box below are male sex accessory ducts that form the part of extra-testicular duct system?

Rete testis, Vas deferens, Epididymis, Urethra, Ureter, Seminiferous tubules

Choose the correct option.
(1) Three (2) Four
(3) Five (4) Two

146. Select the **incorrect** statement w.r.t. humans.
(1) Female sex hormones such as estrogen and progesterone as well as male sex hormones like testosterone can act as GnRH inhibitor.
(2) Ectopic pregnancy is defined as implantation of embryo within uterus at the site other than uterine fundus.
(3) In testis, the regions outside the seminiferous tubules contain immunologically competent cells.
(4) Glans penis is covered by a loose fold of skin called foreskin.

147. The layer of uterine wall that exhibits strong contractions during parturition is characterized by the presence of
(1) Fusiform and unbranched muscle fibres
(2) Striated and multinucleated muscle fibres
(3) Cylindrical and uninucleated muscle fibres
(4) Involuntary and branched muscle fibres

148. Read the given statements and choose the **correct** option w.r.t. humans.
(1) A functional mammary gland is characteristic of all the members of class Mammalia.
(2) The last part of the oviduct, isthmus has the widest lumen and it joins the uterus.
(3) The cavity of the cervix called cervical canal along with uterus forms the birth canal.
(4) Each fallopian tube is about 10-12 cm long and extends from the periphery of each ovary to the uterus.

Space for Rough Work

149. Given below are two statements, one is the **assertion (A)** and the other is **reason (R)**.

Assertion (A): In human males, the testis is extra-abdominal *i.e.*, it is present in a scrotal sac.

Reason (R): The scrotum helps in maintaining the low temperature of the testes (2-2.5°C lower than the normal internal body temperature) necessary for spermatogenesis.

In the light of above statements, select the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

150. Milk is expressed from a structure named 'X' which possesses the opening of

(1) Mammary tubules

(2) Lactiferous ducts

(3) Mammary ducts

(4) Mammary lobes

Goswami Ji

151. Consider the given statements and select the correct option.

Statement A: Spermatogenesis starts at the age of puberty due to significant increase in the secretion of GnRH from adenohypophysis.

Statement B: Oogonial cells in the fetal ovary start division and enter into metaphase-I of the meiotic division and get temporarily arrested at that stage.

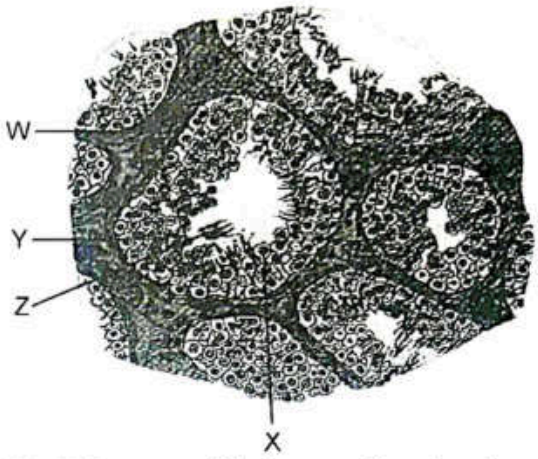
(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

152. Observe the given labelled diagram and choose the **incorrect** option w.r.t. it.



(1) 'W' is responsible for secretion of androgens.

(2) During spermatogenesis, FSH acts on 'X' for the release of certain factors.

(3) 'Z' is formed as the result of a process called spermiogenesis

(4) The ploidy level of 'Y' is same as that of secondary spermatocytes

153. The process of childbirth is called parturition which is induced by a complex neuroendocrine mechanism involving hormones such as

(a) Oxytocin

(b) Cortisol

(c) Estrogen

(d) Prolactin

Select the option with correct set.

(1) (a), (b), (c) and (d) (2) (a), (b) and (c) only

(3) (a), (c) and (d) only (4) (a) and (c) only

154. If a menstrual cycle is of 32 days in a human female, then for how many days luteal phase may persist?

(1) 14 days (2) 18 days

(3) 10 days (4) 20 days

Space for Rough Work

155. Select the **incorrect** match w.r.t. normal pregnancy.

(1)	Chorionic villi	–	Finger-like projections that appear on outer layer of blastocyst
(2)	Trophoblast	–	Gets attached to myometrium
(3)	Inner cell mass	–	Differentiates into ectoderm and endoderm
(4)	Stem cells	–	Have the potency to give rise to all the tissues and organs

156. Identify and select the part of sperm that possesses the structures which produce energy for the movement of tail to facilitate sperm motility essential for fertilization.
(1) Head (2) Neck
(3) Middle piece (4) Acrosome

157. The total number of primary follicles which are left in healthy human ovaries at puberty is about
(1) 1,20,000-1,60,000 (2) 60,000-80,000
(3) 1,00,000-1,20,000 (4) 1,60,000-1,80,000

158. Read the given statements w.r.t. human embryonic development:
(a) By the end of the second month of pregnancy, the foetus develops limbs and digits.
(b) The first movements of the foetus and appearance of hair on the head are usually observed during the fifth month of pregnancy.
(c) After one month of pregnancy, the embryo's heart is formed.
(d) By the end of first trimester, the body of foetus is covered with fine hair, eye-lids separate, and eyelashes are formed.
How many of the above given statements is/are **correct**?
(1) Two (2) Three
(3) One (4) Four

Goswami Ji

159. Given below are two statements w.r.t. humans, one is the **assertion (A)** and the other is **reason (R)**.
Assertion (A): Sex of the baby is determined by the father and not by the mother.
Reason (R): All the haploid gametes produced by the females have the sex chromosome X whereas in the male gametes, fertilising the female gametes, have either X or Y sex chromosome.
In the light of above statements, select the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

160. If due to some reason, the ejaculatory duct in the human reproductive system gets blocked, then the gametes will not be transported from
(1) Rete testis to vasa efferentia
(2) Epididymis to vas deferens
(3) Vasa efferentia to epididymis
(4) Vas deferens to urethra

161. In human testes, the total number of testicular lobules is 'P' while the total number of seminiferous tubules is 'Q'. Identify 'P' and 'Q' respectively and select the correct option.
(1) 250, 250-750 (2) 200, 400-600
(3) 150, 300-500 (4) 500, 500-1500

162. In human males, the secretions of which of the following accessory glands mainly helps in the lubrication of the penis?
(1) Seminal vesicle (2) Bulbourethral gland
(3) Prostate gland (4) Bartholin's gland

163. All of the given components are present in seminal plasma, **except**
(1) Sperms (2) Fructose
(3) Calcium (4) Certain enzymes

164. In human females, menstrual cycle ceases around 50 years of age; that is termed as menopause. This is characterized by
(1) Increase in the level of gonadotropins in the blood
(2) Increase in the level of estrogen in the blood
(3) Decrease in the level of GnRH
(4) Increase in the level of progesterone in plasma

165. Which of the following events is not associated with secretory/luteal phase of menstrual cycle in human females?
(1) Formation of temporary endocrine gland named corpus luteum
(2) No development of new follicles
(3) Maintenance of endometrium of uterus
(4) Peak levels of plasma LH

166. Read the given statements and select the **correct** option w.r.t. human females.
Statement A: Prolactin and estrogen synergise in breast growth but estrogen antagonizes the milk producing effect of prolactin on breast.
Statement B: The breasts enlarge during pregnancy in response to high circulating levels of estrogens, prolactin and progesterone.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

Goswami Ji

167. Choose the **incorrect** option w.r.t. human placenta.
(1) Facilitates the supply of oxygen and nutrients to the embryo
(2) Acts as an endocrine tissue and produces several hormones like estrogen and progesterone
(3) Connected to the embryo through chorionic villi
(4) Assists in removal of carbon dioxide and excretory/waste materials produced by the embryo

168. The synthetic form of which hormone is usually used by doctors to induce parturition?
(1) Prolactin (2) Oxytocin
(3) Thyroxine (4) Cortisol

169. Which of the following statements are correct regarding female reproductive system?
(A) Uterus is supported by ligaments attached to the pelvic wall.
(B) Each ovary is about 2 to 4 cm in length and is connected to the pelvic wall and uterus only by specialised connective tissue.
(C) The shape of the uterus is like an upright pear.
(D) The parts of the female reproductive system along with a pair of the mammary glands are integrated structurally and functionally to support the processes of ovulation, fertilisation, pregnancy, birth and child care.
Choose the most appropriate answer from the options given below.
(1) A and D only (2) A, B and C only
(3) A, C and D only (4) A and B only

170. Select the correct statement w.r.t. humans.
(1) Capacitation occurs prior to fertilization in ejaculatory duct.
(2) Acrosome has a pointed structure used for piercing and penetrating the egg.
(3) LH acts on the Leydig cells and stimulates synthesis and secretion of androgens.
(4) Vas deferens receives a duct from prostate gland and opens directly into urethra.

171. Choose the correct option representing only diploid cells from the following sets.
(1) Spermatogonia, Primary oocyte, Oogonia
(2) Primary spermatocyte, Primary oocyte, Spermatid
(3) Secondary oocyte, Ootid, First polar body
(4) Primary spermatocyte, Ovum, Oogonia

Space for Rough Work

172. Match column I with column II and select the correct option w.r.t. humans.

	Column I		Column II
a.	Primary follicle	(i)	Possesses primary oocyte and a theca layer
b.	Secondary follicle	(ii)	Many degenerate during the phase from birth to puberty
c.	Tertiary follicle	(iii)	Characterised by presence of zona pellucida
d.	Graafian follicle	(iv)	Possesses multiple layers of granulosa cells and an antrum

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iv), b(iii), c(ii), d(i)

173. The total number of mammary lobes present in each breast of human females is
(1) 10-12 (2) 22-25
(3) 15-20 (4) 30-40

174. Select the **incorrect** statement w.r.t. humans.
(1) In humans, the fusion of a sperm with ovum usually occurs in the uterine cavity.
(2) Soon after the infant is delivered, the placenta is also expelled out of the uterus.
(3) The mammary glands of the female starts producing milk towards the end of pregnancy.
(4) A pair of mammary glands is anatomically situated in the upper thoracic region overlaying the musculature.

Goswami Ji

175. Choose the **odd** one w.r.t. the levels of hormones that are increased several folds in the maternal blood during pregnancy.
(1) Prolactin (2) Thyroxine
(3) Gonadotropins (4) Estrogen

176. How many secondary spermatocytes are required to form 600 million spermatozoa in human males?
(1) 300 million (2) 600 million
(3) 200 million (4) 100 million

177. Zona pellucida gets disintegrated
(1) Just prior to implantation
(2) Before rupture of Graafian follicle
(3) After embedding of blastocyst in endometrium
(4) After parturition

178. Select the **correct** statement w.r.t. humans
(1) Lack of menstruation is always an absolute indicator of pregnancy.
(2) In a 28 days menstrual cycle, follicular phase occurs from day 15th to day 25th.
(3) The functions of male sex accessory ducts and glands are maintained by the testicular hormones
(4) The anterior portion of acrosome is covered by an elongated haploid nucleus.

179. Choose the correct match.

(1)	Foetal ejection reflex	–	Induced by under-developed foetus and ovary
(2)	Vitelline membrane	–	Possess receptors for sperm binding
(3)	Infundibulum	–	Funnel shaped part of oviduct closer to uterus
(4)	Spermiation	–	Release of sperms from the seminiferous tubules

180. During follicular phase of menstrual cycle, the levels of which among the following hormones is low throughout the phase?
(1) FSH (2) Estrogen
(3) LH (4) Progesterone

❑ ❑ ❑

Space for Rough Work

For Answers & Solutions

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Goswami Ji



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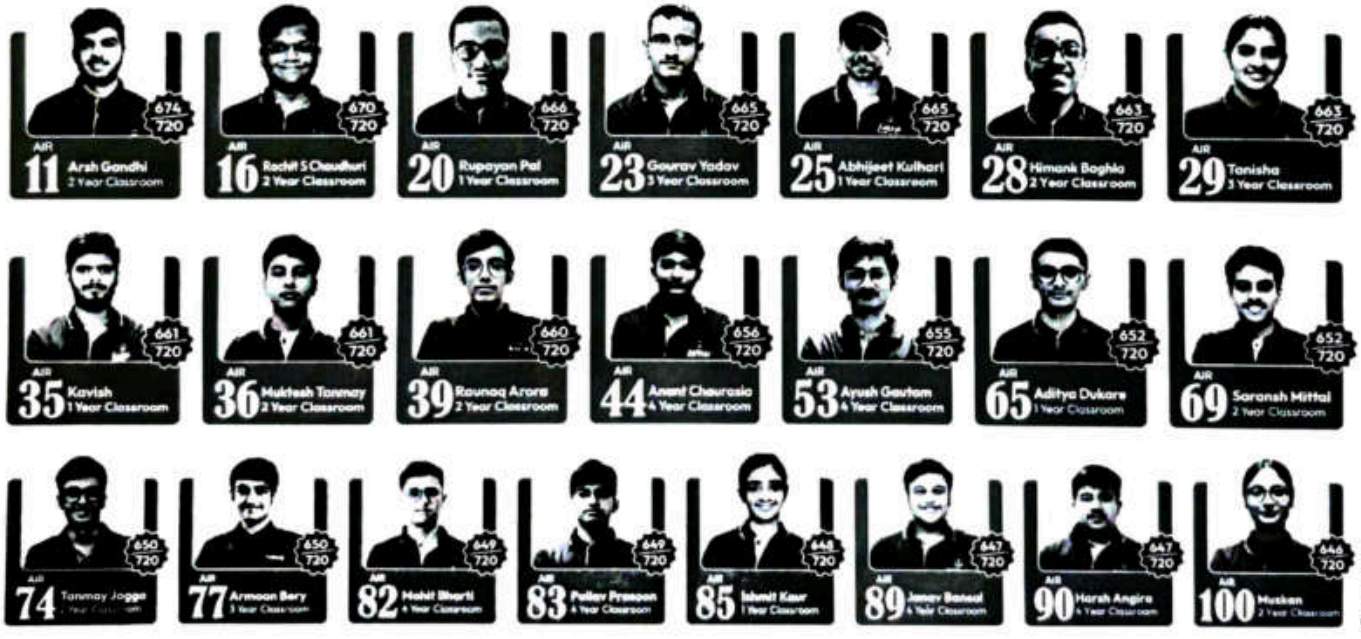
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